LEAP Hand mapping: glove frame → 16 motor angles

haptx_hand_stream (UDP 9871) → leap_teleop.py → RealLeap / leap_hand_node.py all angles in radians

1 Glove input

one UDP frame per glove; for each finger f:

$\theta^f_{mcp},\ \theta^f_{pip},\ \theta^f_{dip},\ \theta^f_{add}$ — IK joint angles (AlternateHandIk), flexion < 0

$q^f_1,\ q^f_2,\ q^f_3$ — joint rotations as unit quaternions (w, x, y, z), each relative to its parent bone

2 Orientation features

where the mapping uses a "feature"

$\Delta_j = r_j^{-1} \otimes q_j$ — rotation since the open-hand rest pose r_j

$\mathbf{v}_j = 2\,\mathrm{atan2}(\|\Delta_{xyz}\|,\ \Delta_w)\ \Delta_{xyz} / \|\Delta_{xyz}\|$ — rotation vector = axis × angle

$\varphi = \mathbf{v}_j \cdot \hat{\mathbf{a}}$ — signed angle about the calibrated axis $\hat{\mathbf{a}}$

3 Joint targets

LEAP joint i = 0…15, sim convention (0 = flat hand)

$$t_i = o_i + \sum_k g_{ik}\, x_k, \qquad x_k \in \{\ \theta,\ \varphi\ \}$$

g_{ik}, o_i from leap_mapping[_right].json

No inputs → $t_i = o_i$ (held). An input missing this frame → t_i keeps its last value.

LEAP finger	motor IDs	motors	driven by your
index	0–3	side, MCP, PIP, DIP	index
middle	4–7	side, MCP, PIP, DIP	middle
ring	8–11	side, MCP, PIP, DIP	pinky (your ring finger is unused)
thumb	12–15	cmc, axl, mcp, ipl	thumb (each joint mixes all 4 features)

Example (right hand): $t_1 = 0.0482 - 2.3034\,\theta^{index}_{mcp}$; a 40° bend, $\theta = -0.698$ → $t_1 = 1.656$ rad (95°)

4 Calibration

leap_calibrate.py: poses p, glove value x_p, LEAP pose angle y_p

$$(g_i,\ o_i) = \arg\min \sum_p (y_p - o_i - g_i\, x_p)^2$$

fingers: one straight line per joint (np.polyfit)

$$W = (\tilde{X}^\top \tilde{X} + \lambda N I)^{-1}\, \tilde{X}^\top (Y - \bar{Y})$$

thumb: ridge fit, $\tilde{X} = (X - \mu)/s$, $\lambda = 0.02$, N = number of poses

$g = W/s, \qquad o = \bar{Y} - g\,\mu$ — back to the raw glove values

5 Smoothing

leap_teleop.py control loop

$$\tilde{t}_i \leftarrow \tilde{t}_i + \alpha\,(t_i - \tilde{t}_i), \qquad \alpha = 1 - 0.35^{\,60\,\Delta t}$$

65 % of the gap per 60 Hz tick; glove silent > 0.5 s → hold the pose

6 Motor command

RealLeap (--real) or leap_hand_node.py (--send)

$$m_i^* = \sigma_i\ \mathrm{clip}(\tilde{t}_i,\ \ell_i,\ u_i) + \pi$$

$[\ell_i, u_i]$ = LEAPsim_limits, $\sigma_i = -1$ for --flip, $+\pi$: the motor reads 180° when the hand is flat

$$m_i \leftarrow m_i + \mathrm{clip}(m_i^* - m_i,\ \ -v\,\Delta t,\ \ v\,\Delta t)$$

speed limit v = 8 rad/s: a jump in the data never snaps the hand

m_i goes to the Dynamixel motors: current-based position control, Kp 600, Kd 200, current cap 550 mA.

Example, continued: $m_1^* = 1.656 + \pi = 4.798$ rad.